

→ it's difficult to compute gradient descent

→ Numerical examples

→ 2 values of

| $x_0$ | $x_1$ | $x_2$ | $y$ |
|-------|-------|-------|-----|
| 1     | 1     | 1     | 3   |
| 1     | 2     | 1     | 4   |
| 1     | 3     | 3     | 5   |

$$n = 3 \quad \theta_0, \theta_1, \theta_2$$

$$d = 2 \quad L = 0 \quad L_2 = 0$$

$$h = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

2nd iteration

apply gradient descent  
compute new value of  $\theta$   
after 2 iterations

Feature  
→  $x_1, x_2, x_3$

parameter  
are →  $\theta_1, \theta_2, \theta_3$

$$\text{Ans) } \theta_0 = \theta_0 - \alpha \sum_{i=1}^n (h_0^{(i)} x_0^{(i)} - y^{(i)}) x_0^{(i)}$$

$$\rightarrow x_0^{(i)} = 1$$

$$\rightarrow \theta_0 = \theta_0 - \alpha \sum_{i=1}^n (h_0 - h_0 y^{(i)}) x_0^{(i)}$$

$$\rightarrow \theta_0 = \theta_0 - \alpha \sum_{i=1}^n (h_0 - y^{(i)}) \cdot 1$$

$$\rightarrow \theta_0 = \theta_0 - \alpha \sum_{i=1}^n (h_0 - y)$$

assuming  $x_0^{(i)} \rightarrow 1$ st feature  
 $y^{(i)} \rightarrow 2$ nd feature

Given  $n=3, \alpha=0.1$

$$\rightarrow \theta_0 = \theta_0 - \alpha \sum_{i=1}^3 (h_0 - y) \cdot 1$$

$$h_0 = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$\text{as } \theta_0 = \theta_1 = \theta_2 = \theta_3 = 0$$

$$h = 0 + 0 + 0 = 0$$

$$\rightarrow \theta_0 = h - y = [(0-3), (0-4), (0-5)]$$

$$= [-3, -4, -5]$$

$$\theta_0 = 0 - 0.1 \sum (h_0 - y) \cdot x_0$$

$$\Rightarrow \theta_0 = 0 - 0.1 \sum (-3, -4, -5) \cdot 1$$

$$\Rightarrow \theta_0 = 0 - 0.1 (-12)$$

$$= 0 + 1.2$$

$$= 1.2$$

$$\theta_1 = 0 - 0.1 \sum ((-3 \times 1) + (-4 \times 2) + (-5 \times 3))$$

$$= 0 - 0.1 \sum (-3 - 8 - 15)$$

$$= 0 - 0.1 \sum (-26)$$

$$\rightarrow \theta_1 = \theta_1 - \alpha$$

$$= 0 -$$

$$= 0 -$$

$$= 0 -$$

$$= 0 -$$

$$= -$$

$$= -$$

$$\rightarrow \theta_2 = \theta_2 - \alpha$$

$$\theta_2 = 0 -$$

$$= -$$

$$= -$$

$$= -$$

$$= -$$

Iterat

1st  
compute  
h<sub>0</sub> values  
using  
formula

Q<sub>0</sub>

h<sub>1</sub>

$$\begin{aligned}
 \rightarrow \theta_1 &= \theta_1^0 - \alpha \sum_{i=1}^3 (h_0(x^{(i)}) - y^{(i)}) x_1^{(i)} \\
 &= 0 - 0.1 \sum_{i=1}^3 (-3, -4, -5) x_1^{(i)} \rightarrow (1, 2, 3) \\
 &= 0 - 0.1 \sum_{i=1}^3 ((-3 \times 1) + (-4 \times 2) + (-5 \times 3)) \\
 &= 0 - 0.1 \sum_{i=1}^3 [(-3 - 8 - 15)] \\
 &= 0 - 0.1 \sum_{i=1}^3 [(-26)] \\
 &= -0.1 \times -26 \\
 &= 2.6
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow \theta_2 &= \theta_2^0 - \alpha \sum_{i=1}^3 (h_0(x^{(i)}) - y^{(i)}) x_2^{(i)} \\
 &= 0 - 0.1 \sum_{i=1}^3 [((-3 \times 1) + (-4 \times 1) + (-5 \times 3))] \\
 &= 0 - 0.1 \sum_{i=1}^3 [-3 - 4 - 15] \\
 &= 0 - 0.1 \sum_{i=1}^3 [-22] \\
 &= -0.1 \times -2.2 \\
 &= 2.2
 \end{aligned}$$

Iteration 1 done  $\rightarrow$  values of  $\theta_0, \theta_1, \theta_2$

let compute  $h_0$  values using formula

| $i$ | $x_1$ | $x_2$ | $h$  |
|-----|-------|-------|------|
| 1   | 1     | 1     | 6.0  |
| 2   | 2     | 1     | 8.6  |
| 3   | 3     | 3     | 15.6 |

$$\begin{aligned}
 h_{\text{est}} &= \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 \\
 &= 1.2 \times 1 + 2.6 \times 1 + 2.2 \times 1 \\
 &= 1.2 + 2.6 + 2.2 = 6.0
 \end{aligned}$$

8.8

$$h_2 = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$= 1.2 \times 1 + 2.6 \times 2 + 0.2 \times 1$$

$$= 1.2 + 5.2 + 0.2 \quad (2.5 \times 2)$$

$$= 8.6 \quad (8.9)$$

$$h_{i=3} = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$= 1.2 \times 1 + 2.6 \times 3 + 2.2 \times 3$$

$$= 1.2 + 7.8 + 6.6$$

$$= 15.6 \quad / 16.5$$

$$h_0 = (6.0, 8.6, 15.6)$$

Iteration 2

$$\theta_0 = 0 - 0.1 \sum_{i=1}^3 [(6-3), (8.6-4), (15.6-5)]$$

$$= 0 - 0.1 \sum [3, 4.6, 10.6]$$

$$\theta_0 = 0 - \alpha \sum_{i=1}^3 (h_0 - y)$$

$$\theta_0 = 0 - \alpha \sum_{i=1}^3 [(6, 8.6, 15.6) - (3, 4, 5)]$$

$$\rightarrow \theta_0 = 1.2 - 0.1 \sum (3, 4.6, 10.6) x_0$$

$$= 1.2 - 0.1 \times 18.2$$

$$= 1.2 - 1.82$$

$$= -0.62$$

$$\theta_1 = 0 - 0.1 \sum_{i=1}^3 (h - y) x_1^{(i)}$$

$$= 2.6 - 0.1 \sum (3 \times (1) + 4.6 \times (2) + 10.6 \times (3))$$

$$= 2.6 - 0.1 \sum (31 - 9.2 - 31.8)$$

$$= 2.6 - 0.1 \sum (-38) = 2.6 - 3.8 = -1.2$$

$$\begin{aligned}\theta_2 &= \theta_2 - 0.1 \sum_{i=1}^3 [(2 \times 0) + (4 \times 1) + (10 \times 6 \times 3)] \\ &= 2.6 - 0.1 \sum [3 + 4 \times 6 + 31.8] \\ &= 2.6 - 0.1 \sum [39.7] \\ &= 2.6 - 3.97 \\ &= -1.37\end{aligned}$$

$$\begin{bmatrix} -0.62 \\ -1.2 \\ -1.37 \end{bmatrix} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ 1 \\ \theta_d \end{bmatrix}$$

GD Algo

Repeat until convergence

$$\theta_j := \theta_j + \alpha \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} \quad (\text{for every } j)$$

| $x_0$ | $x_1$ | $x_2$ | $y$ |
|-------|-------|-------|-----|
| 1     | 1     | 2     | 3   |
|       | 2     | 1     | 4   |
|       | 3     | 3     | 5   |

$h = 3$

$d = 2$

$$(h - y) \rightarrow [-3, -4, -5]$$

$$\theta_2 = 0 - 0.1 \sum$$

$$(-3 \times 2)$$

$$+ (-4 \times 1)$$

$$+ (-5 \times 3)$$

$$= 0 - 0.1 \sum (-6 - 4 - 15)$$

$$= 0 - 0.1 \sum (-25)$$

→ Predicted Labels

Scores  $\geq 0.5 \rightarrow 1$  $\leq 0.5 \rightarrow 0$ Confusion Matrix

| Actual \ Predicted | label | Type |
|--------------------|-------|------|
| 0                  | 1     | FP   |
| 1                  | 1     | TP   |
| 1                  | 0     | FN   |
| 0                  | 0     | FP   |
| 1                  | 1     | TP   |
| 0                  | 1     | FP   |
| 0                  | 1     | FP   |
| 1                  | 1     | TP   |
| 0                  | 1     | FP   |
| 0                  | 0     | FP   |

$$TP = 4 \quad (1, 3, 6, 9)$$

$$FP = 4 \quad (2, 5, 7, 8)$$

$$FN = 1 \quad (4)$$

$$TN = 1 \quad (10)$$

|         |         |
|---------|---------|
| TP<br>4 | FN<br>1 |
| FP<br>4 | TN<br>1 |

$$\rightarrow \text{Accuracy} \rightarrow \frac{TP+TN}{\text{Total}} = \frac{4+1}{10} = 0.5$$

$$\rightarrow \text{Precision} \rightarrow \frac{TP}{TP+FP} = \frac{4}{4+4} = \frac{4}{8} = 0.5$$

Sensitivity, TPR (0.5, 0.5)

$$\hookrightarrow \frac{TP}{TP+FN} = \frac{4}{4+1} = \frac{4}{5} = 0.8$$

$$\text{Specificity} = \frac{TN}{TN+FP}$$

F1 Score

$$\Rightarrow \frac{1}{\frac{1}{4} + \frac{1}{4}} = \frac{1}{1/5} = 0.2$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.5 \times 0.8}{0.5 + 0.8} = \frac{0.8}{1.3} = 0.615$$

ROC Curve  $\rightarrow (TP) \rightarrow 5$   
 $(TN) = 5$

→ Threshold = 0.7

Predicted > 0.7 +ve

[1, 0, 1, 0, 1, 0, 0, 1, 0, 0]

TP = 2 (1, 3)

FP = 2 (5, 8)

TPRate =  $2/5 = 0.4$

FPRate =  $2/5 = 0.4$

→ Threshold = 0.2

TP = 5

FP = 5

TPR = 1.0 FPR = 1

(1.0, 1.0)

Predicted labels

→ [1, 1, 1, 0, 1, 1, 1, 1, 1, 0]

→ CM → TP = 4 Metrics → acc

FP = 4 Accuracy = 0.5

FN = 1 Precision = 0.5

TN = 1 Recall = 0.5

F1 Score = 0.615

ROC Points

Threshold

FPR

TPR

0.7

0.4

0.4

0.5

0.8

0.5

0.4

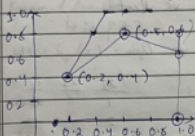
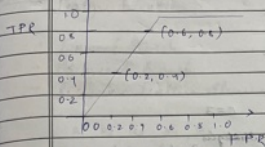
0.8

1.0

0.2

1.0

1.0



$$y = \begin{bmatrix} 1.0 \\ 0.5 \\ 0.0 \end{bmatrix}$$

1.0  
0.5  
0.0

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \quad \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_{nn} \end{bmatrix}$$

← Sample Covariance matrix

$$\begin{bmatrix} u^T & A & u \end{bmatrix}$$

Geiger's decomposition problem

$$x_1 \quad u_1$$

$$x_2 \quad u_2$$

$$x_3 \quad u_3$$

$$x_n \quad u_n$$

$$\rightarrow u^T A u$$

Covariance matrix

9/04/20 Correlation and covariance

| X | Y |
|---|---|
| 1 | 2 |
| 2 | 4 |
| 3 | 5 |
| 4 | 4 |
| 5 | 5 |

Covariance

$$\text{Correlation} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\frac{1}{5} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{5} (1 \cdot 2 + 2 \cdot 4 + 3 \cdot 5 + 4 \cdot 4 + 5 \cdot 5 - 20) = \frac{1}{5} (2 + 8 + 15 + 16 + 25 - 20) = \frac{40}{5} = 8$$

$$\sigma_X = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} = \sqrt{\frac{1}{5} (1^2 + 2^2 + 3^2 + 4^2 + 5^2 - 20)} = \sqrt{\frac{20}{5}} = 2$$

$$\sigma_Y = \sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2} = \sqrt{\frac{1}{5} (2^2 + 4^2 + 5^2 + 4^2 + 5^2 - 20)} = \sqrt{\frac{20}{5}} = 2$$

$$\text{Correlation} = \frac{8}{2 \cdot 2} = 2$$

$$\text{ii. } cov(x, y) = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

$$\text{iii. } \sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

| X | Y | $x - \bar{x}$ | $y - \bar{y}$ |
|---|---|---------------|---------------|
| 1 | 2 | -2            | 4             |
| 2 | 4 | -1            | 0             |
| 3 | 5 | 0             | 1             |
| 4 | 4 | 1             | 0             |
| 5 | 5 | 2             | 1             |

$$\rightarrow cov(x, y) = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

$$= \frac{1}{5} \cdot 44$$

$$= 8.8$$

$$\rightarrow \text{correlation} \rightarrow \frac{cov(x, y)}{\sigma_x \sigma_y} = \frac{8.8}{\sqrt{10} \cdot \sqrt{2}} = 1.911$$

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2 = \frac{1}{5} \cdot 10 = 2$$

$$\sigma_x = \sqrt{2} = 1.414$$

$$\sigma_y = \sqrt{2} = 1.414$$



→ PCA  $x_1, x_2, x_3$   

$$X = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$d_{1,2,3} \rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \rightarrow d_{1,2,3} \rightarrow \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

variance explained by  
through dimension

Step 1: Mean centering  

$$\mu = \begin{bmatrix} 1.5 & 1.5 & 1.5 \end{bmatrix}$$

$$X_c = \begin{bmatrix} 0.5 & -0.5 & -0.5 \\ 1.5 & 0.5 & -0.5 \\ -1.5 & -0.5 & 0.5 \\ -0.5 & -1.5 & 0.5 \end{bmatrix}$$

Step 2: Sample covariance matrix  

$$\begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) & \text{cov}(x_1, x_3) \\ \text{cov}(x_2, x_1) & \text{cov}(x_2, x_2) & \text{cov}(x_2, x_3) \\ \text{cov}(x_3, x_1) & \text{cov}(x_3, x_2) & \text{cov}(x_3, x_3) \end{bmatrix}$$

$$\text{cov} = \frac{1}{n} X_c^T X_c$$

$$= \begin{bmatrix} 1.25 & -0.25 & 0.5 \\ -0.25 & 1.25 & 0.25 \\ 0.5 & 0.25 & 0.25 \end{bmatrix}$$

Step 3: eigenvalues and eigenvectors

$$\lambda_1, \lambda_2, \lambda_3$$

$$\lambda_1 = 0.35, \lambda_2 = 0.35, \lambda_3 = 0.0$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$